

BOEKBESPREKINGS/BOOK REVIEWS

Approximations to the Poisson, Binomial and Hypergeometric distribution functions by *W. Molenaar*.

First edition, 160 pages. Mathematical Centre Tracts No. 31, Mathematical Centre, Amsterdam, 1970. Df. 16,00; U.S.\$5,00.

After having read the article "How to poison Poisson (When approximating binomial trials)" by W. Molenaar in *Statistica Neerlandica*, No. 23 (1969), one will already be quite familiar with the type of approximations that is considered in this book. The author gives a critical discussion of some of the recent publications on the subject. In his study he aims to find approximations of the discrete distributions in such a form that the arithmetical calculations are kept as simple as possible and suitable for calculations on an ordinary desk calculator. He compares the various approximations numerically. It is not always clear whether he succeeds in keeping the calculations fairly simple.

Successively, the author discusses normal approximations to the Poisson distribution; normal and Poisson approximations to the binomial distribution and finally normal, Poisson and binomial approximations to the hypergeometric distribution. It is worth mentioning the usefulness of this result to obtain corrections for continuity for normal approximations of a 2×2 table with fixed marginal totals.

On the whole this is a useful book for those interested in discrete distributions. Although the five pages of relevant literature do contain a good summary on classical and modern research work, the list is by no means complete.

The style is unnecessarily tedious. Besides, the binding of the book in paperback could have been better.

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Probability and Statistical Inference by *R.G. Krutchoff*.

First edition, xiv + 291 pages and 5 tables. Gordon and Breach, Science Publishers, New York, 1970. U.K. £5-5-0; U.S. \$12,60.

The author set himself the task to write a text on probability and statistical inference that should

- (a) be readable at the undergraduate level requiring no statistical prerequisites and only a calculus prerequisite in mathematics,
- (b) treat those topics which are required as preparation for more advanced graduate work,
- (c) contain enough new material to be of interest to researchers and applied statisticians who have already completed their schooling, and
- (d) be detailed enough to be read by those who want to teach themselves without the aid of an instructor.

I think that the author was reasonably successful, except in requirement (a).

The elements of probability theory, from set theory up to Bayes' theorem, are covered in chapter 1 in only 17 pages. It is hard to imagine that a student with no prior knowledge of probability theory will master the topic without much additional reading. Chapters 2 and 3 contain a very satisfactory treatment of 5 discrete and 10 continuous distributions, including the derivation of means, variances and higher order moments in most cases. Some aspects of multivariate distributions are treated in chapter 4. The trinomial and bivariate normal distributions are used for illustration purposes. One wonders why the multinomial distribution is not mentioned. The titles of chapter 5 (moment generating functions) and a short chapter 6 (distributions of functions of random variables) are self-explanatory.

Concepts like mixed and conditional distributions, the weak law of large numbers, the central limit theorem and three types of convergence are grouped together in chapter 7.

Chapter 8 is a short introduction to statistical inference. The distributions of certain sample moments are treated in chapter 9. The author defines, unconventionally, sample

central moments by $\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^r$ and uses the description "sample moments about the average" for $\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^r$. Chapters 10 and 11 contain a reasonable treatment of properties of estimators. Nothing is said about efficiency or about decision functions and related topics, and, with the exception of the method of maximum likelihood, no methods for the determination of estimators are given. The theory of hypothesis testing is presented in chapter 12. The Neyman-Pearson theorem is proved and applied to a number of cases. Much attention is paid to the "randomization" of tests which enables one to make the size of a test equal to a prescribed value. The final chapter contains a discussion of confidence intervals, including Bayes' type and fiducial type confidence intervals.

As a text book this book does not compare favourably with e.g. Mood and Graybill.

The book contains many important and well-known results as well as interesting results not so widely known. One gets the impression that the value of the book could have been enhanced greatly by the inclusion of a small selection of important results that have been omitted, especially in chapter 1 and the chapters on statistical inference. Appendix A, containing a summary (17 pages) of the most important results of the text, is certainly one of the highlights of the book. Appendix 8 contains 5 tables.

There are quite a number of misprints and other inaccuracies in the book. Most of them are (fortunately) quite obvious. The print is clear and the book is durably bound.

Although I cannot recommend this work as a text book, I have no doubt that it deserves its place on one's shelf as a reference book. Besides the very useful appendix A, it contains lots of stimulating results which may be of interest to advanced students and mathematical statisticians.

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Statistical and other tables

First edition, 14 pages and 11 tables. The Institute of Statisticians, London, U.K. 5s.

This is a collection of abridged tables for use of examination candidates, containing the following: ordinates and cumulative distribution of the standardised normal curve, percentage points of the t , χ^2 , F and Poisson distributions, a thousand random numbers, factors for obtaining control chart limits from average range, the z transformation of the correlation coefficient, 5-figure logarithms, 4-figure square roots and a table of the exponential function.

All the tables are standard, the only inconvenience perhaps being that the statistical tables are too much abridged, e.g. the normal distribution which is tabulated over the range 0.00(0.05)3.00(0.10)4.0, and the Poisson distribution for the set of parameter values 0.1, 0.5(0.5)3.0(1.0)15.0. Similarly in the case of the t , χ^2 and F distributions the degrees of freedom increase in steps of one only up to 6 or 10.

There is always need for a convenient set of examination tables, particularly for an undergraduate course in basic statistics. This booklet meets the requirements in many respects, though an extract of the binomial distribution and perhaps confidence limits for proportions would have been valuable additions. It may, however, certainly be recommended.

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Statistische Methoden, ein Soforthelfer

by L. Sachs.

First edition, 103 pages, 25 tables and 5 figures. Springer Verlag, Berlin, 1970. D.M. 8,80; U.S. \$2,50.

This is a collection of formulae and computing instructions for confidence limits and statistical tests in the situations which are most likely to occur in practice.

It is also attempted to explain the most important relevant concepts and to indicate under what circumstances the given methods might be appropriate. Since the space available is that of a thick pocket sized booklet this is a very difficult task. I do not think one can give "Instant Statistics" on a few pages, but the author does succeed in giving concise descriptions. The language is good and most of it can be understood.

I think the text will be useful to the large group of statistically under-educated users of Statistics. It will not be useful to those who have no previous knowledge about Statistics, and to those who have had some substantial training in it. The pure instruction part might be of more general use.

I consider the selection of the material as reasonably good, there is also a handy and useful table appended. There are some mistakes, which I shall communicate to the author. What I dislike about the book is the absence of any material on analysis of variance, surely the most powerful of all statistical methods, and the utter neglect of the all important concept "standard error".

The print is clear but the general layout is not too attractive.

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